

Project Tasks Solutions

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1 Training on Frozen Lake environment using different algorithms.

We have used a 20x20 map to give the feeling of a large map along with a reasonable convergence time.

1.1 Value Iteration (DP approach):

Convergence Time: 53 seconds

The DP approach often fast comes at a computational cost due to the huge amount of states it had to process.

1.2 Monte-Carlo Control (First-visit):

Convergence Time: 186 seconds

Monte-Carlo took the longest of all algorithms but it required least amount of episodes of experience to find the optimal compared to the TD methods during training.

1.3 SARSA:

Convergence Time: 20.32 seconds

The SARSA model converged faster than Q-learning model but it required more episodes than Monte-Carlo and Q-learning to train properly.

It also required epsilon as unity and then be decreased to 0.1 at a steady rate i.e it required more rigorous exploration.

1.4 Q-learning:

Convergence Time: 27.67 seconds

The Q-learning model might have taken slightly more but it was best out of the two model-free algorithms. It had the capability of working at lesser number of episodes as well as smaller exploration.

2 Running the cliffwalking experiment to compare the working of SARSA and Q-learning models.

We ran both the SARSA and q-learning models on the cliffwalking environment. The parameters were:

- episodes = 1000
- map size = 4 x 12
- learning rate = 0.1
- discount factor = 0.9
- rewards = +20 if reached goal and -100 if fell off cliff and a -1 penalty per timestep

Observations: The SARSA model involved the agent following a longer path farther away from the cliff increasing time taken to reach goal but reducing falling to cliff. whereas the q-learning agent followed the optimal path i.e the shortest.

Although both models achieved the goal, it was seen that during the course of training the SARSA model fell down the cliff much less times than the q-learning model. So we tracked the model's result at end of each episode as whether they reached the goal or fell down the cliff. the results:

- SARSA:
Reached Goal: 710
Fell down Cliff: 290
- Q-learning:
Reached Goal: 304
Fell down Cliff: 696

We see that SARSA fell lesser times than the Q-learning model which followed a riskier path.

Conclusion: We conclude that SARSA prioritizes to maximize its rewards over the course of episodes rather act greedily at each step unlike Q-learning. Thus, giving a less riskier path as its optimal path.